

GEETHA BOMMAREDDY

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SUMMARY

AI/ML Engineer with 5+ years of experience in AI/ML — with the last 2+ years focused exclusively on Generative AI — designing, building, and deploying production-grade GenAI systems for large enterprises in financial services. Deep expertise in LLMs (OpenAI GPT-4o, Claude, Gemini), LangChain, Agents, Vector Databases, Prompt Engineering, and Fine-Tuning (LoRA/QLoRA/PEFT), underpinned by a strong software engineering foundation in OOP, data structures, algorithms, and clean coding practices. Proven experience with Python, PyTorch, TensorFlow, NLP, and big data technologies — alongside cloud-native engineering on AWS, Azure, and GCP using microservices, FaaS, and PaaS architectures. Track record at JP Morgan Chase delivering scalable, production GenAI solutions processing 100K+ daily events, leading workstreams independently, and providing clear technical guidance to team members. AWS Certified Data Engineer. Comfortable with DevSecOps, SaaS, SRE, ADO, GitHub, and SonarQube in fast-paced enterprise environments.

SKILLS

Generative AI & LLM Engineering: OpenAI (GPT-4o, GPT-4), Anthropic Claude, Google Gemini, LangChain, LangGraph, LlamaIndex, AI Agents & Multi-Agent Orchestration (AutoGen, CrewAI), Vector Databases (Pinecone, FAISS, Weaviate, pgvector), Prompt Engineering, Fine-Tuning (LoRA/QLoRA/PEFT), RAG Pipelines, Embedding Models, Semantic Search, Hugging Face Transformers

AI/ML Engineering: PyTorch, TensorFlow, Keras, Scikit-learn, NLP (NER, Classification, Summarization, Sentiment Analysis), Supervised & Unsupervised Learning, Anomaly Detection, Time Series (LSTM, ARIMA, Prophet), Ensemble Methods (XGBoost, LightGBM, CatBoost), Explainable AI (SHAP), Model Evaluation & Observability

Software Engineering Foundation: OOP & Design Patterns, Data Structures & Algorithms, Clean Code & Code Instrumentation, Python, SQL/NoSQL, Java (familiar), Golang (familiar), REST APIs, FastAPI, Flask, NodeJS (familiar), React (familiar), Microservices Architecture, Unit & Integration Testing, Code Quality (SonarQube)

DevSecOps & Engineering Practices: DevSecOps, SRE Practices, SaaS & Agile (XP, Lean, Kanban), Azure DevOps (ADO), GitHub & GitHub Actions, SonarQube, CI/CD Pipelines (Jenkins), Docker, Kubernetes (EKS), MLflow, Model Versioning, Drift Detection, Blue-Green Deployments, Automated Testing & Quality Gates

Big Data & Cloud-Native Engineering: PySpark, Apache Spark, Kafka, Airflow, ETL/ELT Pipelines, FaaS & PaaS (AWS Lambda, Azure Functions), Microservices, Batch & Real-Time Processing, Data Modeling, Data Quality & Governance, PostgreSQL, MongoDB, DynamoDB

Cloud & Data Platforms: AWS (SageMaker, S3, Glue, Redshift, Lambda, EC2, EKS), Microsoft Azure (Azure ML, Azure OpenAI, Azure Functions), Google Cloud (Vertex AI, Cloud Functions, BigQuery), Databricks, Snowflake, Multi-Cloud AI Deployment

Certifications: [AWS Certified Data Engineer](#)

EXPERIENCE

JP Morgan Chase & Co. | AI/ML Engineer

Jan 2024 – Present | USA

- Designed and deployed production-grade GenAI systems at JPMC using OpenAI GPT-4, Claude, Gemini, LangChain, and LangGraph — applying deep expertise in Agents, Vector Databases, Prompt Engineering, and fine-tuning (LoRA/QLoRA/PEFT) — delivering scalable, high-quality AI solutions processing 100K+ daily financial transactions with meticulous attention to OOP principles, code quality, instrumentation, and performance standards.
- Led GenAI engineering workstreams independently — managing and prioritizing multiple tasks across fast-paced delivery cycles, providing clear technical guidance to junior engineers, and collaborating with cross-functional teams of data scientists, product managers, and stakeholders — consistently meeting deadlines while maintaining high standards of code quality across 24x7 production systems.
- Engineered end-to-end RAG pipelines and AI Agent systems using LangChain, LangGraph, and Vector Databases (FAISS, Pinecone) — implementing document ingestion, chunking, embedding generation, and semantic retrieval for LLM-powered knowledge assistants — reducing analyst turnaround to same-day insights across regulatory and policy domains with clean, well-instrumented, production-ready code.
- Built cloud-native GenAI solutions on AWS, Azure, and GCP using FaaS (Lambda, Azure Functions), PaaS (SageMaker, Azure ML, Vertex AI), and microservices architectures — applying DevSecOps and SRE practices with GitHub Actions, SonarQube, and CI/CD pipelines — ensuring secure, scalable, and observable AI systems deployed across multi-cloud enterprise environments.
- Applied deep expertise in PyTorch, TensorFlow, Keras, and NLP libraries to develop production AI/ML systems — including transformer-based behavioral models, autoencoders, and ensemble methods (XGBoost, LightGBM)

across 300K+ accounts — demonstrating strong software engineering foundations with OOP design patterns, clean code practices, and rigorous code instrumentation throughout the model lifecycle.

- Built scalable big data and GenAI data pipelines on AWS (S3, Glue, Redshift, Airflow) using PySpark and Kafka — designing cloud-native, microservice-based data infrastructure that supported RAG pipeline embedding generation, real-time feature engineering, and multi-tenant AI workloads — transitioning batch workflows to near real-time processing at enterprise scale.
- Maintained high-quality, well-documented GenAI codebases across 15+ production AI use cases — applying SonarQube code quality standards, unit and integration testing, and comprehensive system documentation — providing clear technical guidance to junior engineers and building sustainable, maintainable AI solutions that minimized technical debt.
- Deployed production GenAI systems following DevSecOps and SAFe methodologies — using Docker, Kubernetes (EKS), Azure DevOps (ADO), and GitHub Actions for automated CI/CD — implementing model versioning, blue-green deployments, and SRE observability practices achieving 60% reduction in inference latency while maintaining enterprise-scale reliability.
- Established AI observability and model monitoring frameworks — applying drift detection, performance tracking, and SHAP-based explainability — ensuring GenAI systems met quality standards and regulatory requirements across compliance, product, and engineering stakeholders in a fast-paced financial services environment.

LTIMindtree | Data Engineer (ML & Analytics)

May 2020 – Jul 2022 | India

- Built production AI systems using PyTorch, autoencoders, and Isolation Forest — applying strong OOP foundations, clean coding practices, and rigorous code instrumentation — identifying 8,200+ anomalous financial events per quarter with meticulous attention to code quality and AML compliance standards.
- Developed NLP pipelines using Hugging Face Transformers, spaCy, and Python — applying text classification, sentiment analysis, and summarization across large volumes of unstructured financial data — reducing processing time from 4 hours to under 30 minutes while improving regulatory reporting accuracy.
- Engineered high-performance ETL pipelines using Apache Spark, AWS Glue, and Airflow — applying Agile and Lean delivery practices, automated deployment pipelines, and SRE observability standards — processing 200K+ daily records while reducing query latency from 120ms to 40ms through systematic database optimization.
- Implemented scalable MLOps pipelines using Docker, Kubernetes, GitHub Actions, and AWS SageMaker — enabling continuous integration, automated deployments, and model lifecycle management — and automated feature engineering workflows using Databricks and MLflow, reducing data preparation efforts by 100+ hours per month.

PROJECTS

Enterprise GenAI Platform — LLMs, Agents, RAG Pipelines & Cloud-Native Deployment

- Independently led engineering of an enterprise GenAI platform using OpenAI GPT-4, Claude, LangChain, LangGraph, and multi-agent orchestration (AutoGen, CrewAI) — designing RAG pipelines, Vector Database infrastructure (FAISS, Pinecone), and cloud-native microservices on AWS and Azure — delivering production AI systems across 1M+ financial records with clean, well-instrumented, and thoroughly documented code.
- Applied DevSecOps, SAFe, SonarQube, GitHub Actions, and Azure DevOps (ADO) to automate deployment pipelines — implementing Prompt Engineering, LoRA/QLoRA fine-tuning, and inference optimization achieving 60% latency reduction — while providing technical guidance to team members across diverse enterprise AI workstreams.

EDUCATION

Master of Science in Information Systems | Trine University, Angola, IN

Dec 2023